

Evidence stack: passive city data and counterfactual TSC networks

Bus cities define the cross-city information budget; RESCO networks provide closed-loop counterfactual control.

Open transit city bundles

Singapore

789 lines
1.24M transitions
observed OD/PV + speed
bands

Austin

234 lines
0.31M transitions
schedule demand/speed
proxies

Halifax

176 lines
0.27M transitions
route APC apportioned

MBTA

940 lines
0.97M transitions
stop board/alight
apportioned

RESCO traffic-signal targets

Regular grids

grid4x4
arterial4x4
880 transitions each

Cologne

cologne1/3/8
60, 180, 480 transitions

Ingolstadt

ingolstadt1/7/21
60, 420, 1260
transitions

Native RL checks

official RESCO 42 runs
LibSignal 36 runs

Passive validation

dynamics + information budget
no action counterfactuals

Counterfactual rollout

same SUMO seeds + phases
closed-loop queue/trip metrics

Claim boundary

TSC is main policy evidence
city logs motivate transfer